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A/D converter and semiconductor device

Abstract

According to one embodiment, an A/D converter includes a successive approximation algorithm setting register that stores a plurality of successive approximation algorithms, an algorithm selection unit that selects a predetermined successive approximation algorithm from the plurality of successive approximation algorithms, a control circuit that generates a comparison value based on the selected predetermined successive approximation algorithm, a DAC that generates a comparison voltage from the comparison value, and a comparator that compares an analog input voltage with the comparison voltage. The control circuit generates a comparison value from a result of the comparison made by the comparator based on the selected predetermined successive approximation algorithm, and converts an analog input voltage into a digital signal from the result of the comparison made by the comparator the number of times equal to the number of bits of the digital signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application claims priority from Japanese Patent Application No. 2022-090982 filed on Jun. 3, 2022, the content of which is hereby incorporated by reference to this application.

BACKGROUND

(2) The present disclosure relates to an A/D converter and a semiconductor device.

(3) Patent document 1 (Japanese Patent Application Laid-open No. 2003-283336) discloses a successive approximation type A/D converter.

SUMMARY

(4) It is desired to improve accuracy of A/D conversion by an A/D converter.

(5) Other problems and novel features will be apparent from the description of the present specification and the accompanied drawings.

(6) According to one embodiment, an A/D converter includes: a successive approximation algorithm setting register storing a plurality of successive approximation algorithms; an algorithm selection unit selecting a predetermined successive approximation algorithm from the plurality of successive approximation algorithms; a control circuit generating a comparison value based on the

selected predetermined successive approximation algorithm; a DAC generating a comparison voltage from the comparison value; and a comparator making a comparison of an analog input voltage and the comparison voltage. The control circuit: generates the comparison value from a result of the comparison made by the comparator based on the selected predetermined successive approximation algorithm; and converses the analog input voltage into the digital signal from the result of the comparison made by the comparator only at least a number of times equal to a number of bits of digital signals.

(7) According to one embodiment, a semiconductor device includes an A/D converter, the A/D converter includes: a successive approximation algorithm setting register storing a plurality of successive approximation algorithms; an algorithm selection unit selecting a predetermined successive approximation algorithm from the plurality of successive approximation algorithms; a control circuit generating a comparison value based on the selected predetermined successive approximation algorithm; a DAC generating a comparison voltage from the comparison value; and a comparator making a comparison of an analog input voltage and a comparison voltage. The control circuit: generates the comparison value from the result of the comparison made by the comparator based on the selected predetermined successive approximation algorithm; and converts the analog input voltage into the digital signal from the result of the comparison made by the comparator at least a number of times equal to a number of bits of the digital signal.

(8) According to the above-mentioned embodiment, an A/D converter and a semiconductor device, which are capable of improving accuracy of A/D conversion, can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block drawing exemplifying an A/D converter according to a first comparative example.

(2) FIG. 2 is a drawing exemplifying an A/D conversion result calculated by a binary search algorithm in the A/D converter according to the first comparative example.

(3) FIG. 3 is a drawing exemplifying an A/D conversion result calculated by a binary search in an A/D converter according to a second comparative example when an erroneous determination exists on the way of comparison.

(4) FIG. 4 is a diagram exemplifying an A/D conversion result calculated by a successive approximation algorithm that corrects an erroneous determination by an inserted redundant cycle in the A/D converter according to the second comparative example.

(5) FIG. 5 is a graph exemplifying A/D conversion accuracy in a clock by a successive approximation algorithm, where a horizontal axis indicates clock frequency and a vertical axis indicates A/D conversion accuracy.

(6) FIG. 6 is a graph exemplifying A/D conversion accuracy in a clock by a successive approximation algorithm, where a horizontal axis indicates clock frequency and a vertical axis indicates A/D conversion accuracy.

(7) FIG. 7 is a block diagram exemplifying a semiconductor device including an A/D converter according to an outline of an embodiment.

(8) FIG. 8 is a graph exemplifying the A/D conversion accuracy when the successive approximation algorithm is switched by the setting clock in the A/D converter according to the outline of the embodiment, where a horizontal axis indicates clock frequency and a vertical axis indicates A/D conversion accuracy.

(9) FIG. 9 is a diagram exemplifying a setting table, which is stored in a storage unit, in a semiconductor device including an A/D converter according to a first embodiment.

(10) FIG. 10 is a block diagram exemplifying a semiconductor device including an A/D converter

according to the first embodiment.

(11) FIG. **11** is a block diagram exemplifying a semiconductor device including an A/D converter according to a second embodiment.

(12) FIG. **12** is a block diagram exemplifying a semiconductor device including an A/D converter according to a modification example of the second embodiment.

(13) FIG. **13** is a block diagram exemplifying a semiconductor device including an A/D converter according to a third embodiment.

DETAILED DESCRIPTION

(14) For clarity of explanation, the following descriptions and drawings are omitted and simplified as appropriate. Moreover, in each drawing, the same elements are denoted by the same reference numerals, and a redundant description thereof will be omitted as necessary.

(15) First, analog/digital converters (referred to as A/D converters) according to first and second comparative examples and a problem(s) discovered by the inventor will be described. Thereafter, an A/D converter and a semiconductor device according to an embodiment(s) will be described. Incidentally, the A/D converters according to the first and second comparative examples and their problems are also included in the scope of the technical concept of the embodiment.

First Comparative Example: Successive Approximation Type A/D Converter

(16) FIG. **1** is a block diagram exemplifying an A/D converter according to a first comparative example. As shown in FIG. **1**, an A/D converter **110** of a comparative example includes a sample hold circuit **111**, a DAC (Digital to Analog Converter) **112**, a comparator **113**, and a control circuit **114**. The control circuit **114** has a successive approximation register **115** and an

addition/subtraction circuit **116**. The A/D converter **110** sequentially compares an analog input voltage V_{in} with a comparison voltage generated by the DAC **112** to output a digital signal D_{out} .

(17) The sample hold circuit **111** holds and samples the inputted analog input voltage V_{in} . The sample hold circuit **111** outputs the sampled analog input voltage V_{in} to the comparator **113**. The DAC **112** converts a comparison value of a digital value to be compared, which is outputted from the control circuit **114**, into an analog value to generate a comparison voltage. The DAC **112** outputs the generated comparison voltage to the comparator **113**.

(18) The comparator **113** compares the analog input voltage V_{in} outputted from the sample hold circuit **111** and the comparison voltage outputted from the DAC **112**. The comparator **113** outputs a comparison result(s) to the control circuit **114**. The control circuit **114** generates a comparison value(s) based on the successive approximation algorithm. Also, the control circuit **114** generates a comparison value from the comparison result of the comparator **113** based on the successive approximation algorithm. Furthermore, the control circuit **114** converts an analog input voltage V_{in} into the digital signal D_{out} based on a result(s) of the comparison made by the number of times corresponding (equal to) to the number of bits of the digital signal D_{out} based on the successive approximation algorithm.

(19) The successive approximation algorithm indicates how the comparison value generated by the control circuit **114** is generated. For example, in a general successive approximation algorithm, a change amount L to be added to or subtracted from the comparison value is sequentially changed every $\frac{1}{2}$ from $\frac{1}{2}$ of the maximum value of the digital signal D_{out} . Such a successive approximation algorithm is called a binary search algorithm.

(20) The addition/subtraction circuit **116** adds/subtracts the change amount L to/from the comparison value based on the comparison result of the comparator **113**. The successive approximation register **115** sequentially outputs a comparison value(s), which is obtained by adding/subtracting the change amount L by the adding/subtracting circuit **116**, to the DAC **112**. Consequently, the DAC **112** successively converts the comparison values into the comparison voltages. Therefore, the comparator **113** compares the analog input voltage V_{in} with the comparison voltage. Such comparison and addition/subtraction of the comparison value are repeated as many times as the number of bits of the digital signal D_{out} .

(21) FIG. 2 is a diagram exemplifying an A/D conversion result, which is calculated by a binary search algorithm, in an A/D converter **110** according to a first comparative example. As shown in FIG. 2, when the A/D converter **110** converts the analog input voltage V_{in} into a 4-bit digital signal D_{out} , the comparator **113** performs comparison of the analog input voltage V_{in} and the comparison voltage $C1$ four times as Comparisons C1 and C2.

(22) First, in first comparison C1, the addition/subtraction circuit **116** adds a change amount $L1$ with respect to an initial value of the comparison value. The change amount $L1$ with respect to the initial value is $\frac{1}{2}$ of the maximum value of the digital signal D_{out} . Therefore, the successive approximation register **115** generates $\frac{1}{2}$ of the maximum value of the digital signal D_{out} as a comparison value. For example, in a case of a 4-bit digital signal D_{out} , the successive approximation register **115** generates a digital value with the most significant bit set to “1”. The DAC **112** converts the comparison value outputted from the successive approximation register **115** into a comparison voltage. The DAC **112** outputs, for example, 8 [V] to the comparator **113** as a comparison voltage. The comparator **113** compares the analog input voltage V_{in} with a comparison voltage of 8 [V]. As shown in the figure, in Comparison C1, the analog input voltage V_{in} is higher. Therefore, the addition/subtraction circuit **116** adds the next change amount $L2$ to the comparison value based on the comparison result of the comparator **113**. The next change amount $L2$ is $\frac{1}{4}$ of the maximum value of the digital signal D_{out} . Also, the successive approximation register **115** determines the most significant bit as “1”.

(23) Next, in second Comparison C2, the successive approximation register **115** generates a comparison value to which the change amount $L2$ is added. For example, in a case of a 4-bit digital signal D_{out} , the successive approximation register **115** generates a digital value in which the second most significant bit is “1”. The DAC **112** converts the comparison value outputted from the successive approximation register **115** into a comparison voltage. The DAC **112** outputs 12 [V], which is obtained by adding (+4) the change amount $L2$, to the comparator **113** as a comparison voltage. A comparator **113** compares the analog input voltage V_{in} with a comparison voltage of 12 [V]. As shown in the figure, in Comparison C2, the analog input voltage V_{in} is smaller. Therefore, the addition/subtraction circuit **116** subtracts the next change amount $L3$ from the comparison value based on the comparison result of the comparator **113**. The next change amount $L3$ is $\frac{1}{8}$ of the maximum value of the digital signal D_{out} . Also, the successive approximation register **115** determines the second most significant bit as “0”.

(24) Next, in third Comparison C3, the successive approximation register **115** generates a comparison value with the change amount $L3$ subtracted. For example, in the case of a 4-bit digital signal D_{out} , the successive approximation register **115** generates a digital value with the third most significant bit set to “1”. The DAC **112** converts the comparison value outputted from the successive approximation register **115** into a comparison voltage. The DAC **112** outputs 10 [V], which is obtained by subtracting (-2) the change amount $L3$, to the comparator **113** as a comparison voltage. The comparator **113** compares the analog input voltage V_{in} with a comparison voltage of 10 [V]. As shown in the figure, in Comparison C3, the analog input voltage V_{in} is higher. Therefore, the addition/subtraction circuit **116** adds the next change amount $L4$ to the comparison value based on the comparison result of the comparator **113** as described above. The next change amount $L4$ is $\frac{1}{16}$ of the maximum value of the digital signal D_{out} . Also, the successive approximation register **115** determines the third most significant bit as “1”.

(25) Next, in fourth Comparison C4, the successive approximation register **115** generates a comparison value to which a change amount $L4$ is added. For example, in the case of a 4-bit digital signal D_{out} , the successive approximation register **115** generates a digital value in which the fourth most significant bit is “1”. The DAC **112** converts the comparison value outputted from the successive approximation register **115** into a comparison voltage. The DAC **112** outputs 11 [V], which is obtained by adding (+1) the change amount $L4$, to the comparator **113** as a comparison voltage. The comparator **113** compares the analog input voltage V_{in} with a comparison voltage of

11 [V]. As shown in the figure, in Comparison C4, the analog input voltage V_{in} is higher.

Therefore, based on the comparison result of the comparator **113**, the successive approximation register **115** determines the fourth most significant bit as “1”. Therefore, the successive approximation register **115** outputs **1011** as the digital signal D_{out} .

(26) Thus, the A/D converter **110** of the first comparative example repeatedly compares the analog input voltage V_{in} held by the sample hold circuit **111** with the comparison voltage generated by the DAC **112** based on the binary search algorithm. Consequently, the analog input voltage V_{in} is converted into the digital value D_{out} . The comparison value to be set in the DAC **112** is set in the successive approximation register **115**.

(27) The binary search algorithm sequentially changes the change amount L , which sequentially changes the comparison value, every $\frac{1}{2}$ from $\frac{1}{2}$ of the maximum value of the digital signal D_{out} in the comparison made for the number of times equal to the number of bits of the digital signal D_{out} . Then, depending on the comparison result of the comparator **113**, the change amount is added or subtracted to or from the comparison value. That is, the binary search algorithm compares the analog input voltage V_{in} with the comparison voltage only the preset number of times equal to the number of bits in the digital signal D_{out} and, in each comparison, the comparison voltage is reduced every predetermined multiple of a base voltage width in a predetermined sequence.

Second Comparative Example: Insertion of Redundant Successive Approximation

(28) Next, an A/D converter according to a second comparative example will be described. A redundant successive approximation is inserted in the A/D converter of this comparative example. FIG. 3 is a drawing exemplifying an A/D conversion result calculated by a binary search in an A/D converter according to a second comparative example when an erroneous determination exists on the way of comparison.

(29) As shown in FIG. 3, the A/D conversion by the binary search algorithm of this comparative example includes erroneous determinations in Comparisons C2 and C3. In Comparison C2, since the analog input voltage V_{in} is smaller than the comparison voltage, the change amount $L3$ should be -2 , but a change amount $L3$ is $+2$. In this way, if there is the erroneous determination in the way of the comparison, an error Error increases after passing through the Comparisons C1 to C4.

(30) FIG. 4 is a diagram exemplifying an A/D conversion result calculated by a successive approximation algorithm that corrects an erroneous determination by an inserted redundant cycle in the A/D converter according to the second comparative example. As shown in FIG. 4, in the successive approximation algorithm of this comparative example, Comparison C3 inserted during the A/D conversion is redundant comparison CR.

(31) In Comparison C2, the comparator **113** compares the analog input voltage V_{in} and the comparison voltage. As shown in the figure, in Comparison C2, the analog input voltage V_{in} is smaller. Therefore, the addition/subtraction circuit **116** originally subtracts the next change amount $L3$ from the comparison value based on the comparison result of the comparator **113**. However, due to the erroneous determination, the addition/subtraction circuit **116** adds the next change amount $L3$.

(32) Meanwhile, in redundant Comparison C3 (CR), the comparator **113** again compares the analog input voltage V_{in} and the comparison voltage. As shown in the figure, the analog input voltage V_{in} is smaller in redundant Comparison C3 (CR). Then, the addition/subtraction circuit **116** subtracts the next change amount $L4 (=L3)$ from the comparison value based on the comparison result of the comparator **113** as described above. Therefore, the erroneous determination can be corrected.

(33) Thus, in the second comparative example, redundant Comparison CR is inserted between Comparison C2 and Comparison C4. Redundant Comparison CR is made the same comparison as the immediately preceding comparison. In this way, by inserting redundant Comparison CR, it is possible to correct the erroneous determination of the comparison in the A/D conversion and improve A/D conversion accuracy. Incidentally, redundant Comparison CR may be inserted not only between the comparisons, but also after a series of comparisons.

(34) An algorithm that inserts such redundant Comparison CR is called a redundant algorithm. Inserted into the redundancy algorithm is the same redundancy comparison as the immediately preceding comparison at least between comparisons only the number of times equal to the number of bits of the digital signal Dout in the binary search algorithm and/or after comparisons only the number of times equal to the number of bits of the digital signal Dout. In the redundancy algorithm, a position of redundant Comparison CR to be inserted may be between Comparison C1 and Comparison C2, between Comparison C2 and Comparison C3, between Comparison C3 and Comparison C4, after Comparison C4, or the like, and is not limited to this. Also, the redundancy algorithm may insert one redundant Comparison CR or multiple redundant Comparisons CR.

(35) (Problem Newly Found by the Inventor)

(36) For example, in the A/D converters according to the first and second comparative examples, when the A/D conversion accuracy is evaluated, the inventor has found that the A/D conversion accuracy deteriorates depending on a frequency of a clock ADCLK that sets operational timing of the A/D converter. Given as this cause is, for example, self-noise generated at a clock ADCLK time having a specific frequency.

(37) FIGS. 5 and 6 are graphs each exemplifying A/D conversion accuracy at a clock ADCLK by a successive approximation algorithm, where a horizontal axis indicates a frequency of the clock ADCLK and a vertical axis indicates A/D conversion accuracy. FIGS. 5 and 6 show each case of performing the A/D conversion by using mutually different successive approximation algorithms.

(38) As shown in FIGS. 5 and 6, switching between the mutually different successive approximation algorithms makes it possible to improve the A/D conversion accuracy at some clock ADCLK frequencies. For example, when the successive approximation algorithm A1 shown in FIG. 5 is used, the A/D conversion accuracy can satisfy specifications of 1.5 [LSB] at a frequency of around 60 [MHz]. However, when the frequency is around 50 [MHz], the A/D conversion accuracy deteriorates more than the specification of 1.5 [LSB].

(39) Meanwhile, as shown in FIG. 6, when the successive approximation algorithm A2 is used, the A/D conversion accuracy can satisfy the specification of 1.5 [LSB] at a frequency of around 50 [MHz]. However, when the clock frequency is around 60 MHz, the A/D conversion accuracy deteriorates more than the specification of 1.5 [LSB].

(40) In this way, changing the settings of the successive approximation algorithms A1 and A2 improves the A/D conversion accuracy at some frequencies of the clock ADCLK, but deteriorates the A/D conversion accuracy at other frequencies of the clock ADCLK. The successive approximation algorithms A1 and A2 change the dependence of the A/D conversion accuracy with respect to the frequency of the clock ADCLK.

Outline of Embodiment: A/D Converter and Semiconductor Device

(41) Next, an A/D converter and a semiconductor device according to an outline of an embodiment will be described. An A/D conversion circuit of the present embodiment has a function of switching the successive approximation algorithm according to the frequency of the clock ADCLK to be set. FIG. 7 is a block diagram exemplifying a semiconductor device including an A/D converter according to an outline of an embodiment. As shown in FIG. 7, a semiconductor device 1 includes an A/D converter 10, a storage unit 20, a processor 30, and a terminal 19. The A/D converter 10 sequentially compares the analog input voltage Vin with the comparison voltage generated by the DAC 12 to output a digital signal Dout.

(42) The A/D converter 10 includes a sample hold circuit 11, a DAC 12, a comparator 13, a control circuit 14, an A/D conversion control register 17, and a data register. The control circuit 14 has a successive approximation register 15. The A/D conversion control register 17 has a successive approximation algorithm setting register 41, a clock setting register 42, an algorithm selection unit 43, an evaluation successive approximation algorithm setting register 44, an evaluation mode setting register 45, and a mode selection unit 46.

(43) The analog input voltage Vin inputted to the semiconductor device 1 is inputted to the sample

hold circuit **11** via the terminal **19**. The sample hold circuit **11** holds and samples the inputted analog input voltage V_{in} . The sample hold circuit **11** outputs the sampled analog input voltage V_{in} to the comparator **13**.

(44) The DAC **12** converts the comparison value of the digital value to be compared, which is outputted from the control circuit **14**, into an analog value to generate a comparison voltage. That is, the DAC **12** generates a comparison voltage from the comparison value. The DAC **12** outputs the generated comparison voltage to comparator **13**.

(45) The comparator **13** compares the analog input voltage V_{in} outputted from the sample hold circuit **11** and the comparison voltage outputted from the DAC **12**. The comparator **13** outputs a comparison result(s) to the control circuit **14**.

(46) The control circuit **14** generates comparison values based on the successive approximation algorithm. Also, the control circuit **14** generates a comparison value from a result of comparison made by the comparator **13** based on the successive approximation algorithm. The control circuit **14** outputs the generated comparison value to the DAC **12**. Furthermore, the control circuit **14** converts the analog input voltage V_{in} into the digital signal D_{out} based on a result of the comparison made by the comparator **13** at least the number of times equal to the number of bits of the digital signal D_{out} . The control circuit **14** outputs the converted digital signal D_{out} via the data register **18**. The data register **18** temporarily holds the digital signal D_{out} and then outputs it to the processor **30**. The digital signal D_{out} is subjected to a desired processing.

(47) For example, the successive approximation register **15** of the control circuit **14** may generate a comparison value based on the successive approximation algorithm and output the generated comparison value to the DAC **12**. Further, the successive approximation register **15** may output the digital signal D_{out} from a result(s) of the successive approximation.

(48) The successive approximation algorithm setting register **41** stores a plurality of successive approximation algorithms. The plurality of successive approximation algorithms may store a plurality of mutually different successive approximation algorithms or may store the same successive approximation algorithm. For example, the successive approximation algorithm setting register **41** may include successive approximation algorithms **A1** and **A2** etc. that improve the A/D conversion accuracy with the clocks ADCLK within a predetermined frequency range.

(49) As an example, the successive approximation algorithm setting register **41** may store a successive approximation algorithm **A1** that improves A/D conversion accuracy in a range of 45 to 49 [MHz], and a successive approximation algorithm that improves A/D conversion accuracy in a range of 40 to 44 [MHz]. The successive approximation algorithm **A1** and the successive approximation algorithm **A2** are different from each other. Specifically, a frequency range of the clock ADCLK in which the successive approximation algorithm **A1** improves the A/D conversion accuracy is different from a frequency range of the clock ADCLK in which the successive approximation algorithm **A2** improves the A/D conversion accuracy.

(50) Further, the successive approximation algorithm setting register **41** may store the successive approximation algorithm **A1** that improves the A/D conversion accuracy in a range of 45 to 49 [MHz] and the successive approximation algorithm **A1** that improves the A/D conversion accuracy in a range of 60 to 64 [MHz]. The successive approximation algorithm setting register **41** may redundantly store the same successive approximation algorithm **A1** that improves the A/D conversion accuracy in two frequency ranges.

(51) The plurality of successive approximation algorithms stored in the successive approximation algorithm setting register **41** may include at least one of a binary search algorithm and a redundancy algorithm. Also, the plurality of successive approximation algorithms may include a plurality of redundant algorithms, and each redundant algorithm may differ from each other in at least one of positions and the number of redundant Comparisons CR.

(52) The clock setting register **42** outputs, for example, frequency information of a clock ADCLK indicating operational timing of the A/D converter **10**. The clock setting register **42** outputs the

frequency information of the clock ADCLK to the algorithm selection unit **43**.

(53) The algorithm selection unit **43** selects a predetermined successive approximation algorithm from the plurality of successive approximation algorithms stored in the successive approximation algorithm setting register **41**. Therefore, the control circuit **14** generates a comparison value based on the predetermined successive approximation algorithm selected by the algorithm selection unit **43**.

(54) The algorithm selection unit **43** may select a predetermined successive approximation algorithm based on the frequency information outputted from the clock setting register **42**. By adopting such a configuration, the A/D converter **10** can automatically switch the successive approximation algorithms depending on the frequency of the clock ADCLK to be set.

(55) The algorithm selection unit **43** outputs the selected successive approximation algorithm to the mode selection unit **46**. Incidentally, when the mode selection unit **46** selects the A/D conversion mode, the algorithm selection unit **43** may output the selected successive approximation algorithm to the control circuit **14** through the mode selection unit **46**.

(56) The evaluation successive approximation algorithm setting register **44** stores an evaluation successive approximation algorithm to be evaluated. Since the evaluation successive approximation algorithm is evaluated over all frequencies of the clock ADCLK used in the A/D converter **10**, it is evaluated without depending on the frequency of the clock ADCLK. That is, the evaluation successive approximation algorithm is evaluated over each frequency associated with each successive approximation algorithm in the plurality of successive approximation algorithms stored in the successive approximation algorithm setting register **41**. The evaluation successive approximation algorithm setting register **44** outputs the evaluation successive approximation algorithm to the mode selection unit **46**.

(57) The evaluation mode setting register **45** outputs evaluation mode setting information, which sets an evaluation mode, to the mode selection unit **46** when evaluating the A/D conversion accuracy.

(58) The mode selection unit **46** selects either the A/D conversion mode or the evaluation mode based on the evaluation mode setting information outputted from the evaluation mode setting register **45**. Specifically, when the mode selection unit **46** receives the evaluation mode setting information from the evaluation mode setting register **45**, it selects the evaluation mode. Therefore, in that case, the mode selection unit **46** selects the evaluation successive approximation algorithm and outputs it to the control circuit **14**. When the evaluation successive approximation algorithm is selected by the mode selection unit **46**, the control circuit **14** generates a comparison value from the comparison result of the comparator **13** based on the selected evaluation successive approximation algorithm.

(59) Meanwhile, when the mode selection unit **46** does not receive the evaluation mode setting information from the evaluation mode setting register **45**, it selects the A/D conversion mode. Therefore, the mode selection unit **46** outputs the successive approximation algorithm, which is outputted from the algorithm selection unit **43**, to the control circuit **14**. The control circuit **14** generates a comparison value from the comparison result of the comparator **13** based on the successive approximation algorithm selected by the algorithm selection unit **43**.

(60) In this manner, the mode selection unit **46** selects either the evaluation successive approximation algorithm or the predetermined successive approximation algorithm selected by the algorithm selection unit **43** based on the evaluation mode setting information outputted from the evaluation mode setting register **45**.

(61) The storage unit **20** is, for example, a non-volatile memory and includes a Read-Only Memory (ROM), a flash memory, a Solid-State Drive (SSD), or other memory technologies, but is not limited to these. The storage unit **20** may include a CD-ROM, a Digital Versatile Disc (DVD), a Blu-ray (registered trademark) disc, or other optical disc storages, magnetic cassettes, magnetic tapes, magnetic disc storages, or other magnetic storage devices.

(62) The processor **30** is, for example, a CPU (Central Processing Unit), but is not limited to this. The processor **30** may include an FPGA (Field-programmable Gate Array), a quantum processor (quantum computer control chip), or the like.

(63) FIG. **8** is a graph exemplifying the A/D conversion accuracy when the successive approximation algorithm is switched by the setting clock in the A/D converter **10** according to the outline of the embodiment, where a horizontal axis indicates clock frequency and a vertical axis indicates A/D conversion accuracy. As shown in FIG. **8**, a frequency range I of the clock ADCLK is set to the predetermined successive approximation algorithm **A1**. In this case, in the frequency range I, the A/D conversion accuracy can satisfy the specification of 1.5 [LSB]. Therefore, deterioration of the A/D conversion accuracy can be suppressed. In a case of a frequency range II of the clock ADCLK, the successive approximation algorithm **A2** is set. In this case, the A/D conversion accuracy can satisfy the specification of 1.5 [LSB] in the frequency range II. Consequently, the deterioration of the A/D conversion accuracy can be suppressed. Therefore, the A/D conversion accuracy satisfies the specification of 1.5 [LSB] over the frequency range shown in the figure, so that the A/D conversion accuracy can be improved.

(64) In this way, according to the present embodiment, switching to the optimum successive approximation algorithm is performed depending on the frequency of the clock ADCLK, so that it is possible to improve the A/D conversion accuracy over the entire frequency range of the clock ADCLK used.

First Embodiment

(65) Next, an A/D converter and a semiconductor device according to a first embodiment will be described. In an A/D converter of this embodiment, a plurality of successive approximation algorithms stored in the successive approximation algorithm setting register **41** are transferred from the storage unit **20**.

(66) FIG. **9** is a diagram exemplifying a setting table, which is stored in a storage unit **20**, in a semiconductor device **1** including an A/D converter **10** according to a first embodiment. As shown in FIG. **9**, the storage unit **20** stores a plurality of successive approximation algorithms. Each successive approximation algorithm is evaluated about the dependence on the frequency of the clock ADCLK in the evaluation mode of the A/D conversion accuracy. Then, each successive approximation algorithm is associated with the optimum frequency of the clock ADCLK according to the evaluation result.

(67) For example, the optimal successive approximation algorithm **A2** that suppresses the deterioration of the A/D conversion accuracy is associated with the clock ADCLK having a frequency range of 40 to 44 [MHz]. Similarly, the successive approximation algorithm **A1** is associated with the clock ADCLK having a frequency range of 45 to 49 [MHz]. The successive approximation algorithm **A3** is associated with the clock ADCLK having a frequency range of 50 to 54 [MHz]. The successive approximation algorithm **A4** is associated with the clock ADCLK having a frequency range of 55 to 59 [MHz]. The successive approximation algorithm **A1** is associated with the clock ADCLK having a frequency range of 60 to 64 [MHz].

(68) Thus, the storage unit **20** stores, by being written in advance, the plurality of successive approximation algorithms and combinations of the respective frequencies associated with each successive approximation algorithm. That is, the storage unit **20** stores the optimum successive approximation algorithm from the evaluation result of the A/D converter **10** in a table form. Incidentally, the storage unit **20** is also called a storage device that functions as a single unit.

(69) FIG. **10** is a block diagram exemplifying a semiconductor device including the A/D converter according to the first embodiment. As shown in FIG. **10**, the plurality of successive approximation algorithms stored in the successive approximation algorithm setting register **41** may be transferred from the storage unit **20**. Therefore, in conjunction with the frequency of the clock ADCLK set by a user(s), it is possible to automatically switch to the successive approximation algorithm precedingly associated as the optimum algorithm for the frequency. This makes it possible to

improve the A/D conversion accuracy.

(70) Further, even if the frequency dependency of the clock ADCLK for improving the A/D conversion accuracy differs depending on products such as the semiconductor device **1** and the A/D converter **10**, the combination of the plurality of optimum successive approximation algorithms from the storage unit **20** for each product is transferred, which can improve the A/D conversion accuracy.

Second Embodiment

(71) Next, an A/D converter and a semiconductor device according to a second embodiment will be described. In an A/D converter of this embodiment, a user evaluates and determines a plurality of successive approximation algorithms stored in the successive approximation algorithm setting register **41** and respective frequencies corresponding thereto.

(72) FIG. **11** is a block diagram exemplifying a semiconductor device including an A/D converter according to the second embodiment. As shown in FIG. **11**, for example, in an A/D converter **10** of a semiconductor device **2**, the user causes the evaluation mode setting register **45** to output the evaluation mode setting information, and sets a mode selection unit **46** to an evaluation mode. This allows the user to determine the optimum combination of the successive approximation algorithms with respect to each frequency of the clock ADCLK in a user's actual usage environment. The user may then determine a table of the successive approximation algorithms as shown in FIG. **9**. The user may operate the processor **30** and cause it to write the determined combination of the successive approximation algorithms to the successive approximation algorithm setting register **41**. The processor **30** writes the combinations of the plurality of successive approximation algorithms and the respective frequencies associated with each successive approximation algorithm to the successive approximation algorithm setting register.

(73) Therefore, in the successive approximation algorithm setting register **41** of the present embodiment, the combinations of the plurality of successive approximation algorithms and the respective frequencies are written by the processor **30** that operates according to user's settings. According to the present embodiment, the successive approximation algorithm of the A/D converter **10** can be adapted to the user's actual usage environment. Therefore, it is possible to apply the optimum successive approximation algorithm in an actual usage environment of users such as customers, and improve the A/D conversion accuracy.

Modification Example

(74) Incidentally, the mode selection unit **46** may switch not only to the A/D conversion mode and the evaluation mode but also to an all-frequency handling mode. An all-frequency handling mode is used regardless of the frequency without making the successive approximation algorithm depend on the frequency of the clock ADCLK as in the evaluation mode. That is, the successive approximation algorithm is used at all the frequencies of the used clock ADCLK.

(75) FIG. **12** is a block diagram exemplifying a semiconductor device including an A/D converter according to a modification example of the second embodiment. As shown in FIG. **12**, a semiconductor device **2a** includes an A/D converter **10a**, a storage unit **20**, a processor **30**, and a terminal **19**. The A/D converter **10a** has a frequency handling successive approximation algorithm setting register **44a** and a mode setting register **45a** instead of the evaluation successive approximation algorithm setting register **44** and the evaluation mode setting register **45** in the A/D converter **10**.

(76) The frequency handling successive approximation algorithm setting register **44a** stores a frequency handling successive approximation algorithm corresponding to all the frequencies of the used clock ADCLK. That is, the frequency handling successive approximation algorithm setting register **44a** stores the frequency handling successive approximation algorithm used over each frequency associated with the plurality of successive approximation algorithms.

(77) The mode setting register **45a** outputs mode setting information, which sets a mode corresponding to all the frequencies, to the frequency handling successive approximation algorithm

regardless of the frequency of the clock ADCLK. The mode selection unit **46a** selects, based on the mode setting information outputted from the mode setting register **45a**, either the frequency handling successive approximation algorithm or the predetermined successive approximation algorithm selected by the algorithm selection unit **43**. When the mode selection unit **46a** selects the frequency handling successive approximation algorithm, the control circuit **14** generates a comparison value from the comparison result of the comparator **13** based on the selected frequency handling successive approximation algorithm.

(78) According to this modification example, it can switch between a mode of the A/D conversion by using the successive approximation algorithm selected depending on the frequency of the clock ADCLK and an A/D conversion mode by using a specific successive approximation algorithm regardless of the frequency of the clock ADCLK. Therefore, switching of the successive approximation algorithm depending on the frequency of the clock ADCLK can be enabled or disabled. Configurations and effects other than this are included in the description of the first embodiment.

Third Embodiment

(79) Next, an A/D converter and a semiconductor device according to a third embodiment will be described. A semiconductor device of this embodiment includes a clock control circuit that controls the frequency of the clock ADCLK. FIG. **13** is a block diagram exemplifying a semiconductor device including an A/D converter according to the third embodiment;

(80) As shown in FIG. **13**, a semiconductor device **3** further includes a clock control circuit **50** in addition to the A/D converter **10b**, the storage unit **20**, the processor **30**, and the terminal **19**. The clock control circuit **50** is also called a clock control device that functions as a single unit. The clock control circuit **50** controls the frequency of the clock ADCLK of the semiconductor device **3** including the A/D converter **10**. The clock control circuit **50** may automatically change the frequency of the clock ADCLK based on conditions set in the semiconductor device **3** in advance.

(81) The A/D converter **10b** has an A/D conversion control register **17a** instead of the A/D conversion control register **17** in the A/D converter **10**. The A/D conversion control register **17a** has a clock control unit **51** instead of the clock setting register **42** in the A/D conversion control register **17a**. The clock control unit **51** detects a change in the frequency of the clock ADCLK, and outputs the frequency information to the algorithm selection unit **43** based on the detected frequency. The algorithm selection unit **43** selects the successive approximation algorithm based on the frequency information outputted from the clock control unit **51**.

(82) According to this embodiment, the A/D converter **10b** has the clock control unit **51**, and so can detect the change in the frequencies of the clock ADCLK even if the frequency of the clock ADCLK changes automatically. Therefore, the algorithm selection unit **43** can switch the successive approximation algorithm according to the change in frequencies. This makes it possible to improve the A/D conversion accuracy. Configurations and effects other than this are included in the descriptions of the first and second embodiments.

(83) The invention made by the present inventor has been specifically described above based on the embodiments, but the present invention is not limited to the above embodiments and, needless to say, can be variously modified without departing from the scope of the invention. Also, each combination of the configurations of the first to third embodiments and the modification examples is within the scope of the technical idea.

Claims

1. An A/D converter comprising: a successive approximation algorithm setting register storing a plurality of successive approximation algorithms; an algorithm selection unit selecting a predetermined successive approximation algorithm from the plurality of successive approximation algorithms; a control circuit generating a comparison value based on the selected predetermined

successive approximation algorithm; a DAC generating a comparison voltage from the comparison value; a comparator making a comparison of an analog input voltage and the comparison voltage; and a clock setting register outputting clock frequency information indicating operational timing, wherein the control circuit: generates the comparison value from a result of the comparison made by the comparator based on the selected predetermined successive approximation algorithm; and converses the analog input voltage into a digital signal from the result of the comparison made by the comparator only at least a number of times equal to a number of bits of digital signals, and wherein the algorithm selection unit selects the predetermined successive approximation algorithm based on the frequency information outputted from the clock setting register.

2. The A/D converter according to claim 1, wherein the plurality of successive approximation algorithms stored in the successive approximation algorithm setting register is written in a storage unit in advance, thereby being transferred from the storage unit that store combinations of the plurality of successive approximation algorithms and respective frequencies associated with the respective successive approximation algorithms.

3. The A/D converter according to claim 1, further comprising: an evaluation successive approximation algorithm setting register storing an evaluation successive approximation algorithm to be evaluated; an evaluation mode setting register outputting evaluation mode setting information that sets an evaluation mode; and a mode selection unit selecting either the evaluation successive approximation algorithm or the predetermined successive approximation algorithm selected by the algorithm selection unit based on evaluation mode setting information outputted from the evaluation mode setting register, wherein the evaluation successive approximation algorithm is evaluated over respective frequencies associated with the plurality of successive approximation algorithms, and wherein when the mode selection unit selects the evaluation successive approximation algorithm, the control circuit selects the comparison value from the result of the comparison of the comparator based on the selected evaluation successive approximation algorithm.

4. The A/D converter according to claim 1, further comprising: a frequency handling successive approximation algorithm setting register storing a frequency handling successive approximation algorithm used over respective frequencies associated with the plurality of successive approximation algorithms; a mode setting register outputting mode setting information that sets a mode; and a mode selection unit selecting either a frequency handling successive approximation algorithm or the predetermined successive approximation algorithm selected by the algorithm selection unit based on the mode setting information outputted from the mode setting register, wherein when the mode selection unit selects the frequency handling successive approximation algorithm, the control circuit generates the comparison value from the result of the comparison of the comparator based on the selected frequency handling successive approximation algorithm.

5. The A/D converter according to claim 1, wherein the successive approximation algorithm setting register writes, by a processor operating according to user's setting, combinations of the plurality of successive approximation algorithms and respective frequencies associated with the respective successive approximation algorithms.

6. The A/D converter according to claim 1, further comprising a clock control unit detecting a change in frequencies of the clock controlled by a clock control device and outputting the frequency information based on the detected frequencies, wherein the algorithm selection unit selects the predetermined successive approximation algorithm based on the frequency information outputted from the clock control unit.

7. The A/D converter according to claim 1, wherein the plurality of successive approximation algorithms include at least one of a binary search algorithm and a redundancy algorithm, in comparisons made the number of times equal to the number of bits of the digital signal, the binary search algorithm sequentially changing a change amount, by which the comparison value is sequentially changed, starting from $\frac{1}{2}$ of a maximum value of the digital signal and halving the

change amount with each subsequent comparison, and the binary search algorithm adding or subtracting the change amount to or from the comparison value according to the result of the comparison of the comparator, and the redundancy algorithm inserting a same redundant comparison as the immediately preceding comparison at least between the comparison and the comparisons performed the number of times equal to the number of bits of the binary search algorithm and/or after the comparison made the number of times equal to the number of bits.

8. The A/D converter according to claim 7, wherein the plurality of successive approximation algorithms include a plurality of redundancy algorithms composed of the redundant algorithm, and wherein each of the redundancy algorithms differs from each other in at least one of a position and a number of the redundant comparison arranged.

9. A semiconductor device comprising an A/D converter, wherein the A/D converter includes: a successive approximation algorithm setting register storing a plurality of successive approximation algorithms; an algorithm selection unit selecting a predetermined successive approximation algorithm from the plurality of successive approximation algorithms; a control circuit generating a comparison value based on the selected predetermined successive approximation algorithm; and a DAC generating a comparison voltage from the comparison value; and a comparator making a comparison of an analog input voltage and a comparison voltage, wherein the control circuit: generates the comparison value from a result of the comparison made by the comparator based on the selected predetermined successive approximation algorithm; and converts the analog input voltage into a digital signal from the result of the comparison made by the comparator at least a number of times equal to a number of bits of the digital signal, wherein the A/D converter further has a clock setting register that outputs clock frequency information indicating operational timing, and wherein the algorithm selection unit selects the predetermined successive approximation algorithm based on the frequency information outputted from the clock setting register.

10. The semiconductor device according to claim 9, further comprising a storage unit storing, by being written in advance, combinations of the plurality of successive approximation algorithms and respective frequencies associated with the plurality of successive approximation algorithms, wherein the plurality of successive approximation algorithms stored in the successive approximation algorithm setting register are transferred from the storage unit.

11. The semiconductor device according to claim 9, wherein the A/D converter further includes: an evaluation successive approximation algorithm setting register storing an evaluation successive approximation algorithm to be evaluated; an evaluation mode setting register outputting evaluation mode setting information that sets an evaluation mode; and a mode selection unit selecting either the evaluation successive approximation algorithm or the predetermined successive approximation algorithm selected by the algorithm selection unit based on the evaluation mode setting information outputted from the evaluation mode setting register, wherein the evaluation successive approximation algorithm is evaluated over each of frequencies associated with the plurality of successive approximation algorithms, and wherein when the mode selection unit selects the evaluation successive approximation algorithm, the control circuit generates the comparison value from the result of the comparison of the comparator based on the selected evaluation successive approximation algorithm.

12. The semiconductor device according to claim 9, wherein the A/D converter further has: a frequency handling successive approximation algorithm setting register storing frequency handling successive approximation algorithms used across each of the frequencies associated with the plurality of successive approximation algorithms; a mode setting register outputting mode setting information that sets a mode; and a mode selection unit selecting either the frequency handling successive approximation algorithm or the predetermined successive approximation algorithm selected by the algorithm selection unit based on the mode setting information outputted from the mode setting register, and wherein when the mode selection unit selects the frequency handling successive approximation algorithm, the control circuit generates the comparison value from the

result of the comparison of the comparator based on the selected frequency handling successive approximation algorithm.

13. The semiconductor device according to claim 9, further comprising a processor operated at user's settings, wherein the processor writes, to the successive approximation algorithm setting register, combinations of the plurality of successive approximation algorithms and the respective frequencies associated with the respective successive approximation algorithms to the successive approximation algorithm setting register.

14. The semiconductor device according to claim 9, further comprising a clock control circuit controlling a frequency of the clock, wherein the A/D converter further has a clock control unit detecting a change in the frequency of the clock and outputs the frequency information based on the detected frequency, and wherein the algorithm selection unit selects the predetermined successive approximation algorithm based on the frequency information outputted from the clock control unit.

15. The semiconductor device according to claim 9, wherein the plurality of successive approximation algorithms include at least one of a binary search algorithm and a redundancy algorithm, the binary search algorithm sequentially changing a change amount, which sequentially changes the comparison value, starting from $\frac{1}{2}$ of a maximum value of the digital signal and halving the change amount with each subsequent comparison, and the binary search algorithm adding or subtracting the change amount to or from the comparison value, and the redundant algorithm inserting a same redundant comparison as the immediately preceding comparison at least between the comparison and the comparisons performed the number of times equal to the number of bits and after the comparisons performed the number of times equal to the number of bits in the binary search algorithm.

16. The semiconductor device according to claim 15, wherein the plurality of successive approximation algorithms include a plurality of redundancy algorithms composed of the redundant algorithm, and wherein each of the redundancy algorithms differs from each other in at least one of a position and a number of the redundant comparison arranged.
